Taking out the common factor $2, 2, x, y$, we get

$$= 2 \times 2 \times x \times y(x+2)$$
$$= 4xy(x+2)$$

Type 2 : Factorisation by taking out the common binomial factor from each term

Example 3.19

(i) Factorise: $(2x+5)(x-y)+(4y)(x-y)$	(ii) Factorise $3n(p-2)+4(2-p)$
Solution:	***Solution:***
We have $(2x+5)(x-y)+(4y)(x-y)$	We have $3n(p-2)+4(2-p)$ (taking – as common)
Taking out the common binomial factor $(x-y)$	$3n(p-2)-4(p-2)$
we get, $(x-y)(2x+5+4y)$	Taking out the common binomial factor $(p-2)$
	we get, $(p-2)(3n-4)$

Type 3 : Factorisation by grouping

Sometimes, the terms of a given expression are grouped suitably in such a way that they have a common factor so that the factorisation is easy to take out common factor from those terms.

Example 3.20

Factorise : $x^2 + yz + xy + xz$

Solution:

We have, $x^2 + yz + xy + xz$

Group the terms suitably as, $= (x^2 + xy) + (yz + xz)$

$= x(x+y) + z(y+x)$

$= x(x+y) + z(x+y)$ (addition is commutative)

$= (x+y)(x+z)$ [taking out the common factor $(x+y)$]

Type 4 : Factorisation using identities

(i) $(a+b)^2 = a^2 + 2ab + b^2$ (ii) $(a-b)^2 = a^2 - 2ab + b^2$ (iii) $a^2 - b^2 = (a+b)(a-b)$

Example 3.21

Factorise : $x^2 + 8x + 16$

Solution:

Now, $x^2 + 8x + 16$ can be written as $x^2 + 8x + 4^2$

Comparing this with $a^2 + 2ab + b^2 = (a+b)^2$ we get $a = x$; $b = 4$

$$(x^2) + 2(x)(4) + (4)^2 = (x+4)^2$$
$$x^2 + 8x + 16 = (x+4)^2$$

$(x+4)$, $(x+4)$ are the two factors.

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